Srihaas Gunda

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.47
Alphores junior College	Class XII (TSBIE)	2024	97.40%
Alphores high school	Class X (SSC)	2022	10.00%

RELEVANT COURSES

Math Courses: Machine Learning,Probability & random processes, Matrix theory, Complex variables, Differential equations and Transform Techniques.

Computer Science Courses: Introduction to programming, DSA,DBMS

Lab Courses: Electrical Circuits Lab, Digital Systems Lab, Electronic Devices and Circuits Lab, DSP Lab, PCB Design and Layout for electronic systems

Core Courses: Analog electronics, Digital Systems, Circuits and network analysis, Signal Processing, Semiconductor devices and fundamentals, Control systems, Engineering Electromagnetics Electrical machines, Power system analysis

SKILLS

Programming Languages: C, C++, Python

Frameworks & Libraries: NumPy, Matplotlib, pandas, scikit-learn

Tools: MATLAB, Arduino IDE, Git/GitHub, LTspice, Altium Designer, LaTeX, Linux, Quartus Prime Lite

Hardware Description Languages: Verilog

PROJECTS

Industrial Surface Defect Classification using Machine Learning

- Developed a multiclass machine learning pipeline for industrial surface defect classification, comparing Decision Tree, Random Forest, and SVM models on the UCI Steel Plates Faults dataset.
- Optimized a Random Forest using GridSearchCV with 5-fold cross-validation and class balancing, achieving 77% test accuracy and 0.80 Macro F1 while improving minority-class recall.

Movie Sentiment Analysis

- Developed an end-to-end NLP pipeline for sentiment classification using TF-IDF feature extraction and Logistic Regression on the IMDB movie review dataset.
- Performed text preprocessing, hyperparameter tuning, and model evaluation, achieving **91% test accuracy**, and deployed the classifier using Streamlit.

Personalized Fitness Routine Recommender | GitHub

- Developed a personalized fitness routine recommendation system using K-Nearest Neighbors (KNN) to recommend exercises based on fitness goals, experience level, available equipment, workout preferences, and time constraints. Implemented data preprocessing, eligibility-based filtering, and personalized generation of exercise duration, sets, reps, and rest time. Built and deployed an interactive multi-step Streamlit web application for generating customized workout routines.

Breast Cancer Classification using Logistic Regression

- Built a Logistic Regression model to classify breast tumors as benign or malignant using standardized numerical diagnostic features.
- Performed hyperparameter tuning using GridSearchCV with 5-fold cross-validation and developed reusable prediction pipelines for single-sample and CSV-based batch inference.

Traffic Signal Controller (Verilog)

- Designed and implemented an FSM-based traffic signal controller in Verilog to manage traffic flow through timed state transitions.
- Simulated and verified the controller for normal operation, pedestrian crossing requests, and emergency reset conditions.

Eigenvalue Solver using Jacobi Method

- Developed a numerical linear algebra application in C implementing the Jacobi Rotation Algorithm to compute eigenvalues of symmetric matrices.
- Designed iterative pivot selection, orthogonal matrix transformations, and convergence checks to progressively diagonalize matrices with configurable precision.

POSITIONS OF RESPONSIBILITY

Elan and nvision Events head-Organising workshops, competitions and institute-wide events for IIT Hyderabad's annual techno-cultural festival.

Elan and nvision Events & nVision Coordinator - Played a major role in organizing and executing nVision tech summit - 1st edition

SPICMACAY Coordinator - Organized and managed events promoting Indian classical music and heritage, handling logistics, coordination with artists, and audience engagement.